

**PUBLISHED**

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks    |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(b)(ii) | <p>any <b>three</b> from:</p> <ol style="list-style-type: none"> <li>1 (PCR) amplifies / makes many copies of, DNA (fragments) ;</li> <li>2 (PCR) <i>ref. to</i> selecting fragments, small enough / less than 1000 bp / not more than 916 bp (for sequencing) ;</li> <li>3 (electrophoresis) separate the (DNA) fragments ;</li> <li>4 (electrophoresis) confirms / identifies, fragment lengths / fragment number / PCR worked correctly ;</li> </ol>                                                                                                                                                                                               | <b>3</b> |
| 5(c)     | $(2^{35} =) 3.4 \times 10^{10}$ ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>1</b> |
| 5(d)     | <p>any <b>four</b> from:</p> <ol style="list-style-type: none"> <li>1 child may have disease when, neither parent has the disease / no family history ;</li> <li>2 early, detection / treatment (for gene therapy to work) ;</li> <li>3 improve, quality of life / life expectancy ;</li> <li>4 <i>ref. to</i> informs family planning (for future pregnancies) ;</li> <li>5 <i>idea that</i> it does not involve additional sampling ;</li> <li>6 (screening is) expensive ;</li> <li>7 treatment / gene therapy, expensive ;</li> <li>8 test may not always be accurate ;</li> <li>9 long-term effect of treatment, unknown / negative ;</li> </ol> | <b>4</b> |